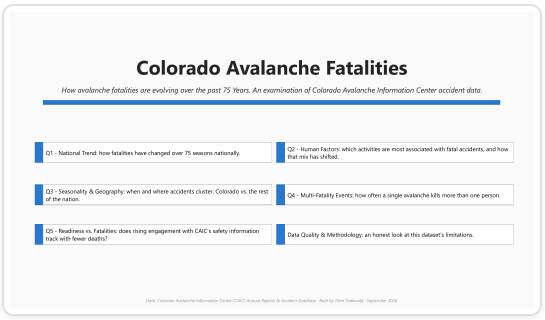
DATA ANALYST PORTFOLIO — PROJECT 2

Colorado Avalanche Fatalities Dashboard

75 years of Colorado Avalanche Information Center accident and engagement data (1951–2025), analyzed end-to-end in Power BI, DAX, and Python.

Built to test a real question — has better gear, forecasting, and education actually reduced avalanche deaths? — using CAIC's public accident database plus a second dataset hand-assembled from ten years of CAIC's own annual reports, since no clean public dataset of backcountry participation exists. Chosen deliberately for its imperfections: investigator judgment calls, shifting record-keeping standards, inconsistent categories, all worked through and documented rather than smoothed over.



Dashboard overview · 7 pages total

1,016	10	2
fatal events analyzed	sources re-verified	bugs found & fixed

SKILLS DEMONSTRATED

Data Modeling & DAX

Built a full Power BI data model (fact + dimension tables, calculated columns, DAX measures) from a 1,016-row dataset. Diagnosed and fixed two distinct bugs: a silent `BLANK()`-vs-zero measure error and a text-vs-number type mismatch breaking a numeric comparison.

Statistical Analysis & QA

Independently cross-checked every reported statistic in Python before publishing it — a habit that caught a real correlation-coefficient error mid-project and led to a transparent, documented correction.

Stakeholder Communication

Resolved an undocumented data ambiguity by emailing the data-owning organization directly, then incorporated their answer, including an unprompted reliability caveat, into the project's documentation.

Python & Data Engineering

Wrote Python/pandas scripts to clean and transform raw accident records into an analysis-ready fact table, collapsing 24 raw activity codes into 6 categories and building three dimension tables from scratch.

Primary-Source Research

Hand-assembled a 10-year metrics dataset from ten separate annual-report PDFs with inconsistent definitions, then re-located and verified all ten against a government archive for long-term citation reliability.

Dashboard & Data Viz Design

Designed a custom, colorblind-validated Power BI theme and a 7-page analytical narrative, then published it publicly via Power BI's Publish to Web for no-login sharing.

THE ANALYSIS

Q1 Trend	Fatalities rose 5x from the 1950s to 1990s, then flattened; 2020–21 is the deadliest season on record.
Q2 Human Factors	Motorized backcountry recreation grew from ~0% to over 40% of fatalities since the 1990s.
Q3 Geography	Colorado and the rest of the nation both peak Feb–March; Colorado runs hotter into March.
Q4 Multi-Fatality	16% of accidents involve multiple deaths but account for 32% of all fatalities.
Q5 Readiness	Safety-info engagement trended up while fatalities showed almost no trend over the same decade.

Live dashboard: app.powerbi.com/view

Repository: github.com/tadewald-trent/avalanche_fatalities_dashboard

Power BI · DAX · Python · pandas · Data Cleaning · Git / GitHub

Trent Tadewald

Data: Colorado Avalanche Information Center · 2026